Designing to the Bottleneck, Not the FLOPs: CNN Image Denoising on an Input-Bandwidth-Bound Small-FPGA Integer Accelerator

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Abstract—Efficient neural networks are usually designed to minimize multiplications, but on a small accelerator the run time may be set by something else. On our integer CNN core for a Zynq XC7Z020 FPGA (64 multipliers, 7-bit inputs), we show how the measured bottleneck changes the design of a Gaussian denoiser. A cycle model, corrected on earlier board runs and within 0.003% of the board on four networks, shows that this build, fed host-built windows one value per cycle, is bound by its input port: a window of K values costs K cycles for up to 64 output channels, so a DnCNN-style baseline leaves most multipliers idle. Two choices that look wasteful by FLOP count therefore pay off. A 2×2 space-to-depth input fits a 6.4× larger network in 25% fewer cycles. Doubling the hidden outputs and pooling them in pairs (maxout) doubles the parameters for 1% more cycles; at the same training budget it denoises better than the ReLU network of the same width, and in one seed averaging the pairs did as well, pointing to the free capacity rather than the max [SEEDS] [AVG]. With quantization-aware training that reproduces the board's arithmetic, the quality model reaches 30.12 dB on Set12 (σ = 25), above BM3D on the same input, and the speed model matches BM3D in 2.63× fewer cycles than the baseline; four networks were verified bit for bit on the board, a full image each. The cycle model predicts that with two input values per cycle pooling stops being free, and with four or with line buffers the 6.4× larger network becomes slower than the baseline: the network to choose follows the measured bottleneck.

Keywords—image denoising, CNN accelerator, FPGA, hardware-aware network design, maxout, quantization-aware training

I. INTRODUCTION

CNN denoisers such as DnCNN [1] outperform the classical BM3D [2] but need tens of thousands of multiplications per pixel, and networks for an accelerator are usually chosen by their multiplication or parameter count. That count is known not to set the run time. The roofline model [3] explains why kernels of low arithmetic intensity are bound by bandwidth rather than compute; ShuffleNet V2 [4] derived design rules from measured GPU speed; EfficientNet-X [5] analysed the gap between FLOPs and latency on TPUs and GPUs and used space-to-depth to raise accelerator utilization; and ABPN [6] built an INT8 super-resolution network from operator latencies measured on a mobile NPU. FPGA CNN accelerators remove memory-bandwidth bottlenecks with on-chip pipelines and line buffers [7]; eCNN [8] co-designed restoration networks with a DRAM-bandwidth-limited accelerator; and light CNN denoisers have been put on Zynq-class FPGAs [9], [10] and on a vendor DPU with INT8 quantization-aware training [11]. These works measure GPU, TPU, NPU or DPU latency or model DRAM traffic. Closest to ours, HALP [12] prunes channels to a GPU's measured latency steps, and DRACO [13] reshapes memory-bound networks to fill a fixed systolic array.

We study the same principle at a smaller scale and at cycle level, measured on the board rather than simulated, on an integer CNN core of our own design for a Zynq XC7Z020, a test vehicle, slower than a PC CPU (Section V): 64 multipliers, one bitstream that runs every layer with only its shape changed, 7-bit unsigned inputs and a ReLU on every output. Because its input port takes one value per cycle, a window costs the number of values streamed, K, and output channels up to the multiplier count cost nothing. Our contributions are:

1. a cycle model, validated on the board, that separates what is free on this core (output channels up to 64, kernel shape, resampling, host-side operations) from what costs (K); 2. an iso-cycle comparison of two choices that this model favours although they look wasteful by FLOP count: space-to-depth, which fits a 6.4× larger network in 25% fewer cycles, and doubled hidden outputs pooled in pairs (maxout), whose hidden layers keep their cycle count on the board, with a mean-pooling control [SEEDS] [AVG]; 3. integer models, trained with quantization-aware training that reproduces the board's arithmetic, that beat BM3D on the same input, verified bit for bit on the board.

Space-to-depth [14], [5], maxout [15]–[17] and quantization-aware training [18] are known techniques; our claim concerns the reason for choosing them. It holds for an input structure that receives host-built windows; with on-chip

line buffers the same model predicts different choices (Section II).

II. Accelerator and Cycle Model

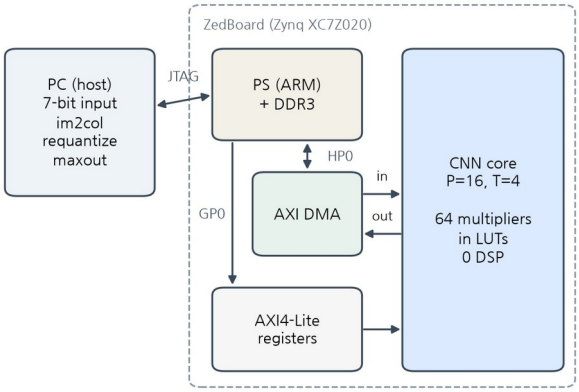


Fig. 1. System. The PC builds the windows (im2col) and requantizes between layers; the core in the PL computes every convolution.

Core. The core receives one window of K activations and computes ReLU(b + Σ w·x) for COUT output channels, with INT8 weights, INT32 accumulation and 7-bit activations. Its 64 multipliers are P = 16 output-channel lanes times T = 4 tap banks; K ≤ 1152 and COUT ≤ 128 are set per run, and the same bitstream also runs a CIFAR-10 classifier. The design uses 21% of the LUTs, 51% of the BRAM and, by Vivado's choice in an unrestricted synthesis, no DSP slice. Timing closes at 100 MHz (WNS +0.221 ns); the board runs use the 50 MHz default clock, and a 100 MHz run gave the same cycle count (0.44 s of core time).

Cycle model. The input port accepts one value per cycle; output channels are processed P at a time in ⌈K/T⌉ cycles per group while the next window loads. A window therefore costs

$$C = \max(\, K, \lceil COUT/P \rceil \cdot \lceil K/T \rceil \,). \quad (1)$$

When K dominates, the core is input-bound and up to P·T = 64 output channels cost nothing: at K = 288, COUT = 32 and 64 both take 288 cycles. The board showed one cycle more per window (on the CIFAR-10 run and the baseline) and at least COUT cycles per window, as outputs leave one per cycle (settled by B's first layer, K = 36, COUT = 64: 64 cycles). With these two parameter-free corrections, (1) matches all four networks within 0.003% (Table II), against 0.4–1.6% without them; the ReLU network, run after they were fixed, was predicted within 0.0014%. All board runs used the sparse mode, which skips multiplications by zero activations; it shortens only the compute term of (1), so input-bound layers take the same cycles.

Where the bottleneck comes from. One value per cycle is a choice of the AXI-Stream wrapper, which unpacks four 7-bit values per 32-bit beat, made to reuse the CIFAR-10 bitstream unchanged. With w values per cycle, about P·T/w output channels come free (64, 32 and 16 for w = 1, 2, 4). Equation (1) is the dense-mode bound: by it, maxout costs 1.9× the ReLU network already at w = 2; sparse mode, estimated with the measured activation densities, leaves 1.4–1.8× and keeps B ahead of A until w = 4. With on-chip line buffers [7], a window brings only one new pixel, C_in values; by the dense model a hidden layer with K = 288 then costs 145 cycles at COUT = 32 and 289 at COUT = 64, maxout doubles the cost, and the baseline (833 cycles/pixel) becomes faster than network B (1,209), in the sparse estimate too. These are predictions, not measurements, and the rules below hold for this input structure. Host im2col, requantization and pooling take 0.07–0.13 s per image on a 4-core PC; on hidden layers requantization and pooling need fewer operations per window than K and could hide in hardware, while im2col would need line buffers.

III. Networks Designed to the Bottleneck

Baseline (A). A 7-layer, 32-channel 3×3 network derived from DnCNN, with BatchNorm folded. Inputs are clipped to [0, 1] and quantized to 7 bits in training and test. Every output passes a ReLU, so the network cannot predict signed noise as DnCNN does; the last layer outputs the clean image and receives the noisy image as an extra input. A's hidden layers (K = 288, COUT = 32) need 144 cycles of compute but 288 of input, so its multipliers are 41% utilized and more multipliers would not make it faster.

Space-to-depth (B). As in FFDNet [14], the image is split into its four 2×2 phases and processed at half resolution. With a quarter of the windows, a 10-layer, 64-channel network fits; its hidden layers (K = 576, COUT = 64) balance input and compute exactly, and the network is 88% utilized. It has 6.4× A's parameters and 1.6× its multiplications in 25% fewer cycles, so its gain over A is that of a larger network that space-to-depth makes affordable, not of the resampling alone.

Free output channels pooled in pairs (B-32 + maxout). At 32 channels, B's hidden layers return to K = 288 and up to 64 outputs are free again. Each hidden layer computes two candidates per channel, and the host keeps the larger after requantization,

$$\max(\mathrm{ReLU}(a), \mathrm{ReLU}(b)) = \mathrm{ReLU}(\max(a, b)), \quad (2)$$

which is maxout [15] in its Max-Feature-Map form [19]. The next layer's K is unchanged, so hidden layers keep their cycle count; only the first layer (K = 36 < 64) grows to 64 cycles per window, 1% per pixel. The pooling makes the free outputs usable: passing all 64 on would double the next layer's K and cycles, and a ReLU network with the same parameters (45 channels, 150k) would cost, by the cycle model, 934 cycles per pixel, 38% more (it was not trained). As a control, an averaging network pools each pair by its mean, with the same parameters and cycles. In restoration the benefit of maxout depends on width: it halved the filters at equal quality in super-resolution [16] and was about 0.04 dB below ReLU in a 64-channel space-to-depth denoiser [17].

Training and quantization. We train on 380 Train400 [1] images, holding out every 20th for validation, with 40×40 (A) or 50×50 patches, flips, rotations, MSE and Adam (10^{-3}, batch 128), halving the rate after 6 epochs without validation gain.

Weights are per-channel symmetric INT8 and activations 7-bit with one scale per map [18], chosen from percentiles on training images. A module that reproduces the board's integer arithmetic in float (rounding, bias units, skip scaling, pooling, 8-bit output) is then fine-tuned for 15 epochs with straight-through gradients and learned scales [20]; its PSNR matches the integer model within 0.001 dB. No checkpoint, learning-rate step or quantization range was chosen on Set12 or BSD68.

IV. Verification

The RTL was verified in a UVM [21] environment (protocol agents, a predicting scoreboard, coverage and assertions) on Verilator: 39 runs under UVM 1.2 and 26 under IEEE 1800.2-2020 passed, and all ten injected bugs were caught. Shape coverage was 89–100% on a reduced build and 17–29% on the board build; the denoisers' hidden shapes are among its directed CIFAR-10 shapes, and their first and last layers were verified on the board only. On the board every layer's output is compared value by value with the integer model; every board figure below passed.

V. Results

Setup. Set12 and BSD68 [22] at σ = 25, with the same noise for every method, clipped and quantized to 7 bits. This input is the CNNs' training distribution but not BM3D's: on unclipped noise BM3D gains 0.03 / 0.26 dB (Set12 / BSD68), and tuning its σ changes it by under 0.1 dB, so Table I lists both. As a CNN reference on the same input we evaluate the public DnCNN-S weights [1] (float, trained on unclipped noise). SSIM [23] ranks our integer models as PSNR does. Integer PSNR is that of the integer model, which the board reproduces bit for bit; with one training seed, differences under 0.1 dB are not interpreted.

TABLE I. MEAN PSNR (DB), SSIM OF THE 8-BIT OUTPUT AND CORE COST. CYCLES ARE THE CORRECTED MODEL; "BOARD" MARKS NETWORKS WHOSE CYCLES WERE MEASURED. B-32 ROWS TRAIN FOR A FIXED 120 EPOCHS (SEED 1).

Model	Cycles/px	Util.	Params	Set12 float / int	BSD68 float / int	SSIM Set12 / BSD68
BM3D [2], clipped input	—	—	—	29.95	28.37	0.850 / 0.799
BM3D, unclipped	—	—	—	29.98	28.62	0.851 / 0.804
DnCNN-S [1], same input	9.3k	—	556k	30.35	28.89	0.859 / 0.822
A: 7 × 32	1,775 board	41%	47k	29.87 / 29.83	28.83 / 28.80	0.842 / 0.809
B + QAT: s2d, 10 × 64	1,323 board	88%	300k	30.13 / **30.12**	29.01 / **29.02**	**0.855** / **0.821**
B-32 + maxout + QAT	675 board	87%	152k	30.00 / **30.00**	28.95 / **28.94**	**0.851** / **0.817**
B-32 ReLU + QAT	669 board	45%	76k	29.58 / 29.62	28.66 / 28.68	0.840 / 0.807
B-32 averaging + QAT [AVG]	675	87%	152k	—	—	—

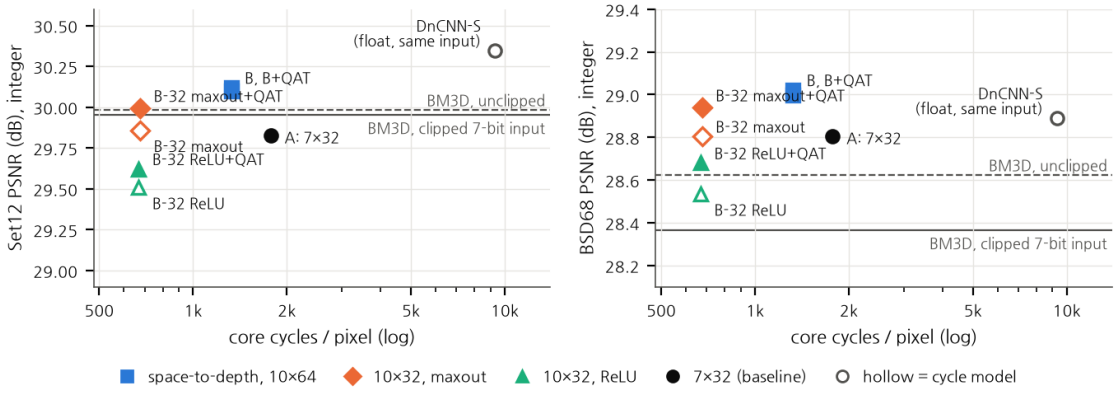


Fig. 2. Integer PSNR against core cycles per pixel (left Set12, right BSD68). Filled markers are board-measured cycles; B-32 points are seed 1.

Quality model. B + QAT reaches 30.12 / 29.02 dB: +0.17 / +0.66 dB over BM3D on the same input, +0.14 / +0.40 dB over unclipped BM3D, and +0.29 / +0.22 dB over A on all 80 images, in 1.34× fewer cycles.

Speed model. B-32 + maxout + QAT needs 2.63× fewer cycles than A and scores 30.00 / 28.94 dB, +0.17 / +0.14 dB above A: level with BM3D on Set12 in PSNR (9/12 images) and SSIM, and +0.57 dB on BSD68 (+0.32 dB unclipped).

Free channels against ReLU, and max against mean. At the same budget, the maxout network scores +0.37 / +0.26 dB (SSIM +0.011 / +0.011) above the ReLU network on all 80 images for 1.01× the board cycles (seed 1; image-paired standard error 0.04 / 0.02 dB, so the training run is the uncertainty). The averaging control matched maxout (validation +0.02 dB in float [AVG]), so in this seed the max adds nothing beyond the extra outputs. The ReLU run collapsed once, at epoch 26, and its validation-driven schedule halved the rate four times while it recovered; before that the three networks were level, so seeds 2 and 3 decide whether the gap holds and how large it is [SEEDS].

Quantization and the same-input reference. QAT brought every integer model within 0.04 dB of its float model (maxout had lost 0.16 / 0.14 dB). On the same input DnCNN-S (556k, float) beats our 300k integer model by 0.2 dB on Set12 and in SSIM on both sets (by 0.001 on BSD68); its BSD68 PSNR deficit reflects training on unclipped noise.

Board. The maxout layers (K = 288, 64 outputs) took 289 cycles per window, exactly as the 32-output layers of A and of the ReLU network: doubling the outputs cost nothing on the board (Table II). Absolute throughput is low: 6.4 GOPS at most at 50 MHz and 0.9–2.3 s of core time per 256×256 image, against 30–38 ms for float PyTorch on a 4-core PC CPU. The JTAG test harness dominates wall-clock time, and the on-chip ARM was not compared; this paper concerns choosing networks for a given core.

TABLE II. BOARD RUNS (SET12 05.PNG, 256×256, 50 MHZ, NO MISMATCH IN ANY LAYER). CYCLES PER WINDOW: MODEL → BOARD; B-32 NETWORKS WITH QAT; THREE EARLIER VERSIONS ALSO MATCHED.

Model	Hidden K / COUT	Cycles/window	PSNR
A	288 / 32	288 → 289	29.83
B + QAT	576 / 64	576 → 577	30.06
B-32 maxout	288 / **64**	288 → **289**	29.94
B-32 ReLU	288 / 32	288 → 289	29.54

VI. CONCLUSION

On a 64-multiplier integer core, a cycle model that matches the board showed that the bottleneck is the input port, not the multipliers, and two choices that look wasteful by FLOP count paid off: space-to-depth fitted a 6.4× larger network into 25% fewer cycles, and doubling the hidden outputs and pooling them in pairs doubled the parameters for 1% more cycles and denoised better, by max and, in one seed, by mean alike [SEEDS] [AVG]. These rules belong to an input structure fed by host-built windows, but the test behind them should carry over: by the cycle model about P·T/w output channels are free, so pooling 64 outputs would again be free on a core of this organization with 256 multipliers fed four values per cycle, and would not be on this core at two. EfficientNet-X [5], HALP [12] and DRACO [13], whose fastest network depends on the array size, apply the same logic on TPUs, GPUs and systolic arrays. Whether P·T/w holds for other input structures is a prediction, to be measured next on wider ports and line buffers. Pooling is free in cycles only: in sparse mode it issued 3.0× the ReLU network's nonzero multiplications. Energy, end-to-end time without the test harness, other noise levels and color images are not reported. Code, models, per-image PSNR and board logs are in the repository.

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